Ipamorelin vs CJC-1295

By Dr. Logan 4 months ago

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CJC-1295 and ipamorelin are both growth hormone secretagogues, though how they naturally raise growth hormone levels differs. Despite their differences in mechanism of action, these two peptides share a lot in common. That's not to say that their effects are identical by any means though. For the discerning researcher, knowing the differences between these two peptides is critical to experimental design and so it is important to explore precisely how they differ. Additionally, the differences in their mechanisms of action mean that these two peptides might be synergistic in certain research applications and so those potential synergies are worth knowing.

Ipamorelin vs CJC-1295: Mechanism of Action

Ipamorelin is a growth hormone secretagogue receptor (GHS-R) agonist, which means it mimics the effects of ghrelin. Ipamorelin was derived from GHRP-1, which was itself derived as one of six (6) different molecules from ghrelin. As an agonist of the GHS-R, ipamorelin activates cells in the anterior pituitary gland to cause growth hormone release. It also binds to GHS-R subtypes in other areas of the brain to regulate reward cognition, learning, memory, the sleep-wake cycle, taste sensation, and glucose metabolism.

CJC-1295 is a derivative of growth hormone releasing hormone (GHRH) and binds to the GHRH-R in the anterior pituitary gland. As a GHRH-R agonist, CJC-1295 has fewer direct effects than ipamorelin because there is no GHRH-R outside of the anterior pituitary gland. Thus, CJC-1295 primarily causes an increase in growth hormone levels. The increase in growth hormone, however, has secondary effects that will be discussed in greater detail.

Ipamorelin vs CJC-1295: Structure

Ipamorelin is a much smaller molecule when compared to CJC-1295, but CJC-1295 offers the ability to be modified easily through the addition of a molecule known as DAC. DAC, or the drug affinity complex, can drastically alter the half-life of CJC-1295 and thus offers an easy means of tailoring the activity of the peptide. Both peptides must be administered Sub-Q for optimal results as there is extensive degradation in the GI tract as well as first-pass metabolism in the liver.

Ipamorelin

Peptide Sequence: Aib-His-D-2Nal-D-Phe-Lys

Molecular Formula: $C_{38}H_{49}N_9O_5$

Molecular Weight: 711.868 g/mol

PubChem CID: 9831659

CAS Number: 170851-70-4

Source: **PubChem**

CJC-1295 Structure

Sequence: Tyr-D-Ala-Asp-Ala-Ile-Phe-Thr-Gln-Ser-Tyr-Arg-Lys-Val-Leu-Ala-Gln-Leu-Ser-Ala-Arg-Lys-Leu-Leu-Gln-Asp-Ile-Leu-Ser-Arg

Molecular Formula: $C_{152}H_{252}N_{44}O_{42}$

Molecular Weight: 3367.954 g/mol

PubChem CID: 56841945

Source: **Oxford Academic**

CJC-1295-DAC Structure

Sequence: Tyr-D-Ala-Asp-Ala-Ile-Phe-Thr-Gln-Ser-Tyr-Arg-Lys-Val-Leu-Ala-Gln-Leu-Ser-Ala-Arg-Lys-Leu-Leu-Gln-Asp-Ile-Leu-Ser-Arg-**Lys-DAC**

Molecular Formula: $C_{165}H_{269}N_{47}O_{46}$

Molecular Weight: 3647.954 g/mol

PubChem CID: 91976842

Source: **Oxford Academic**

Ipamorelin vs CJC-1295: Administration

A brief word on administration of these two peptides is necessary. While CJC-1295 is strictly administered via subcutaneous direction, ipamorelin can be administered IV or subcutaneously. This makes ipamorelin slightly more flexible for administration and opens the ability to tailor infusion rates if experimental design is necessary. Additionally, IV administration makes for a faster rise and fall of ipamorelin levels and can help to minimize first pass effects depending on the location of administration.

Ipamorelin vs CJC-1295: Growth Hormone Release

When it comes to growth hormone release, CJC-1295 and ipamorelin could not be more different. The half-life of **ipamorelin** is in the order of 2 hours whereas the half-life of CJC-1295 (with DAC) is on the order of days. In fact, research shows that it may be as long as 8 days. And that is just for a single dose. When given as multiple doses, CJC-1295 can cause IGF-1 levels to remain elevated for as long as a month[1].

Their vastly different half-lives make ipamorelin and CJC-1295 drastically different in their effects on growth hormone. Ipamorelin causes a very prominent spike in growth hormone release followed by a return to baseline relatively quickly. CJC-1295, on the other hand, causes a less prominent, more prolonged increase in growth hormone that lasts for days.

It isn't just the length of the growth hormone spike that differs between these two peptides though, it is also the pattern of growth hormone release the differs. Under normal physiological circumstances, growth hormone release varies throughout the day based on circadian rhythm, energy consumption and expenditure, sleep, and cues about growth. This normal ebb and flow pattern of growth hormone release is preserved by CJC-1295, which can be thought of as shifting the range of values in which growth hormone is secreted, but not the pattern. Ipamorelin has a much more dramatic effect on both the range as well as the pattern of growth hormone release. However, the effects of ipamorelin wear off quickly. Thus, neither peptide has a dramatic long-term effect on growth hormone secretion patterns, but ipamorelin does have an acute impact. This can be used to achieve different results based on when ipamorelin is administered in regard to activities like exercise, eating, sleep, and so forth.

Ipamorelin vs CJC-1295: Weight Loss

CJC-1295 was one of the first peptides investigated for the treatment of lipodystrophy, or abnormal fat deposition. That mantle that has now been taken up by drugs like tesamorelin and semaglutide, but that doesn't mean research into fat-burning peptides has been abandoned. Quite the opposite is true, in fact, and CJC-1295 has become of renewed interest for its fat-burning properties. Ipamorelin, which also has robust fat-burning properties, has been shown to stimulate the loss of up to 14% of fat mass in animal models over 12 months.

Ultimately, the fat-burning properties of **CJC-1295** have not been quantified, but it is reasonable to speculate that they would likely be more robust than those of ipamorelin. There are two reasons for this. First, CJC-1295 has a prolonged half-life and thus a prolonged effect on growth hormone levels. This is almost guaranteed to lead to more substantial changes in body composition.

The second reason to speculate that CJ-1295 would provide more effective weight loss is that tesamorelin, which is also a GHRH-R agonist shows slightly better results in weight loss than ipamorelin. In fact, tesamorelin is associated with a 15% decrease in fat mass over a single year. Tesamorelin has a half-life of just ~40 minutes. It is likely that CJC-1295, with its similar mechanism of action and significantly prolonged half-life, would have a much more significant effect on body composition than tesamorelin, putting it well ahead of ipamorelin. Given the recent emphasis on weight loss drugs like semaglutide by the public, it is a good time to design research studies to reevaluate the ability of CJC-1295 to induce weight loss.

Ipamorelin vs CJC-1295: Body Composition

Burning fat mass is not the only thing that growth hormone is good at. It is also excellent at increasing lean body mass. This primarily means increasing both muscle mass and bone mass. CJC-1295, as the most potent GHRH analogue on this list has very robust benefits for body composition and particularly fat mass. Ipamorelin is no slouch though and, when combined with the right diet can produce very prominent changes in muscle mass. Where ipamorelin really shines, however, is in the promotion of bone growth. Research shows that ipamorelin, because of its substantial direct effects on GHS receptors in locations other than the central nervous system, can increase bone formation by stimulating both mineralization of existing bone and the deposition of new bone[2]. The benefits of ipamorelin on bone health are particularly prominent in the setting of chronic wasting conditions that arise due to disease or because of certain medications like glucocorticoids (e.g., prednisone). For instance, ipamorelin completely stops bone loss in the setting of long-term glucocorticoid use and increases bone formation by as much as four-fold[3]. It has been suggested that it might be prudent to administer ipamorelin in conjunction with glucocorticoids both to offset their effects at standard doses and to allow for increased steroid dosing in difficult-to-manage conditions.

Ipamorelin vs CJC-1295: Infertility

Early research on CJC-1295 shows that it may promote ovulation in infertile female patients. It has been well-established that IGF-1 signaling is a critical component of ovulation and that certain infertility conditions, like polycystic ovarian syndrome, are mediated to some degree by changes in IGF-1 signaling. Of course, IGF-1 signaling is dependent to a large degree on growth hormone signaling and thus CJC-1295 was tested for its effect on ovulation. As it turns out, CJC-1295 specifically increases follicular IGF-1 levels via an unidentified mechanism. This leads to enhanced ovulation[4]. While it stands to reason that ipamorelin may also influence ovulation and fertility given its ability to raise growth hormone and thus IGF-1 levels, research to this effect has not been performed. It would be interesting to see if the difference in half-life between these two peptides has an impact as superovulation has not been reported with most growth hormone secretagogues and CJC-1295 has the longest half-life of the known secretagogues.

Ipamorelin vs CJC-1295: Pain Perception

One interesting property of ipamorelin that is not shared by CJC-1295 is its ability to regulate neuropeptide Y. Neuropeptide Y is important in mitigating pain perception, particularly when that pain is neurological or gastrointestinal in nature. Research in rats indicates that ipamorelin may reduce pain perception by as much as 2-fold, cutting the experience of pain by a factor of four[5]. This function stems directly from the fact that there is ghrelin (a.k.a. GHS) receptors located throughout the central nervous system and the GI tract. Ghrelin has long been known to mitigate pain, so it should come as little surprise that a derivative of ghrelin has similar properties. Still, there is much research being performed to understand just how, exactly, ipamorelin mitigates pain perception so that a deeper understanding of pain signaling can be gained.

Ipamorelin vs CJC-1295: Summary

Ipamorelin and CJC-1295 are like one another in many ways and both have prominent effects on growth hormone levels. The quality of their effects differs, however, with ipamorelin causing short, large increases in growth hormone levels and CJC-1295 causing prolonged, less dramatic increases in growth hormone levels. Additionally, because ipamorelin targets the GHS-R and these receptors are found in locations other than the anterior pituitary, ipamorelin has a lot of direct effects in addition to simply stimulating growth hormone release. These properties may be important depending on the nature of the research being undertaken.

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